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LASER DEBOND PROCESS FOR FABRICATION OF HIGH-DENSITY ORGANIC INTERPOSERS

Abstract

Embodiments disclosed herein comprise a method for assembling an interposer. In an embodiment, the method comprises assembling a structure over a carrier substrate, where the structure is mechanically coupled to the carrier substrate by a debond film. In an embodiment, the structure comprises an organic dielectric layer, and the debond film comprises an inorganic layer. The method may further comprise ablating at least a portion of the debond film with a laser. In an embodiment, a wavelength of the laser is within an infrared range. The method may further comprise separating the carrier substrate from the structure.

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Background/Summary

BACKGROUND

[0001] Advanced packaging process flows often employ temporary carriers for assembly processes. These carriers are typically glass wafers with an organic adhesive and release layer that is decomposed thermally or by UV laser absorption when the overlying substrate needs to be released from the carrier. However, such solutions have a high total thickness variation (TTV) (e.g., up to approximately 5 μm or larger). The glass wafers are also not compatible with many silicon wafer handling tools present within a fabrication facility.

[0002] Another approach for carrier solutions is to use a silicon carrier wafer. In such an embodiment, the bond between the wafer and the overlying interposer (or other structure) is not deactivated. Instead, the silicon wafer is thinned with a grinding process that can ultimately remove the entire silicon carrier wafer. This is an expensive process because carriers cannot be reused. Additionally, there is a risk of damage or degradation to the overlying structure during the recessing process. Another solution may include the use of a thermal release bonding film. However, the temperature limit for many interposer solutions is less than approximately 200° C. Thermal bond layers also have a high thickness without a high degree of uniformity (i.e., they may have a high TTV).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1A is a cross-sectional illustration of a carrier with a debond film over the carrier, in accordance with an embodiment.

[0004] FIG. 1B is a cross-sectional illustration of a carrier with a multi-layer debond film, in accordance with an embodiment.

[0005] FIG. 1C is a cross-sectional illustration of a carrier with a multi-layer debond film, in accordance with an embodiment.

[0006] FIGS. 2A-2D are cross-sectional illustrations depicting a process for assembling an interposer on a carrier and releasing the carrier with a laser debond process, in accordance with an embodiment.

[0007] FIG. 3 is a process flow diagram of a process for debonding an interposer from a carrier with a laser process, in accordance with an embodiment.

[0008] FIGS. 4A-4F are cross-sectional illustrations depicting a process for assembling an interposer with dies that are bridged together by a routing layer and releasing the interposer from a carrier with a laser process, in accordance with an embodiment.

[0009] FIGS. 5A-5L are cross-sectional illustrations depicting a process for forming a zero-misaligned via over a carrier, and releasing the interposer from the carrier with a laser process, in accordance with an embodiment.

[0010] FIGS. 6A-6D are cross-sectional illustrations depicting a process for assembling an interposer with dies that are bridged together by a routing layer and releasing the interposer from a carrier with a laser process, in accordance with an embodiment.

[0011] FIGS. 7A-7B are cross-sectional illustrations depicting a process of forming an interposer with multiple layers of dies coupled to a carrier and releasing the carrier with a laser process, in accordance with an embodiment.

[0012] FIGS. 8A-8B are cross-sectional illustrations depicting a process of forming an interposer with conductive pillars and releasing the interposer from a carrier with a laser process, in accordance with an embodiment.

[0013] FIG. 9 is a schematic of a computing device built in accordance with an embodiment.

EMBODIMENTS OF THE PRESENT DISCLOSURE

[0014] Described herein are electronic systems, and more particularly, organic interposer devices that are fabricated through the use of a laser debonding process, in accordance with various embodiments. In the following description, various aspects of the illustrative implementations will be described using terms commonly employed by those skilled in the art to convey the substance of their work to others skilled in the art. However, it will be apparent to those skilled in the art that the present disclosure may be practiced with only some of the described aspects. For purposes of explanation, specific numbers, materials and configurations are set forth in order to provide a thorough understanding of the illustrative implementations. However, it will be apparent to one skilled in the art that the present disclosure may be practiced without the specific details. In other instances, well-known features are omitted or simplified in order not to obscure the illustrative implementations.

[0015] Various operations will be described as multiple discrete operations, in turn, in a manner that is most helpful in understanding the present disclosure, however, the order of description should not be construed to imply that these operations are necessarily order dependent. In particular, these operations need not be performed in the order of presentation.

[0016] Various embodiments or aspects of the disclosure are described herein. In some implementations, the different embodiments are practiced separately. However, embodiments are not limited to embodiments being practiced in isolation. For example, two or more different embodiments can be combined together in order to be practiced as a single device, process, structure, or the like. The entirety of various embodiments can be combined together in some instances. In other instances, portions of a first embodiment can be combined with portions of one or more different embodiments. For example, a portion of a first embodiment can be combined with a portion of a second embodiment, or a portion of a first embodiment can be combined with a portion of a second embodiment and a portion of a third embodiment.

[0017] Future scaling to higher density routing layers and interconnects, has led to an interest in silicon based carriers. Compared to glass-based carriers, silicon carriers offer a lower total thickness variation (TTV), better geometric stability, and compatibility with all silicon wafer handling tools and processes. Further, elimination of the organic adhesive and release layers allows for improved TTV at the point of the adhesive. Improved release layer design can also provide improved thermal and/or mechanical stability that leads to opportunities for high density 2.5D and 3D integration.

[0018] More generally, embodiments disclosed herein include the use of a release layer (also referred to herein as a debond film or a debond layer), that is sensitive to infrared (IR) radiation. In some embodiments, the debond film absorbs the IR radiation so that the debond film is ablated. This allows for the overlying interposer (or other structure) to be released. Silicon wafers are transparent to IR radiation, so the laser can be directed to the debond film through the thickness of the silicon carrier wafer. IR radiation may refer to radiation with a wavelength between approximately 800 nm and approximately 1.0 mm. In a particular embodiment, the IR radiation may have a wavelength between approximately 1.0 μm and approximately 10 μm , or between approximately 1.0 μm and approximately 20 μm . As used herein, “approximately” may refer to a range of values that are within ten percent of the stated value. For example, approximately 10 μm may refer to a range between 9 μm and 11 μm .

[0019] In an embodiment, the debond film may have a material composition suitable for deposition with a well-controlled deposition process, such as physical vapor deposition (PVD), atomic layer deposition (ALD), or chemical vapor deposition (CVD). These deposition processes may enable the formation of thin and low TTV materials. Between the low TTV of the silicon carrier and the low TTV of the debond film, a total TTV of the carrier system may be approximately 5 μm or less, or approximately 1 μm or less. In an embodiment, the debond film may have a thickness that is up

to approximately 5 nm, up to approximately 10 nm, up to approximately 20 nm, up to approximately 30 nm, up to approximately 50 nm, or up to approximately 100 nm. The debond film may have a single layer composition, or a multi-layer composition (as will be described in greater detail below). In the case of multi-layer approaches, the total thickness of all layers of the debond film may be up to approximately 500 nm, up to approximately 600 nm, or up to approximately 1,000 nm. Though, thicker debond films may also be used in some embodiments. [0020] The debond film and silicon carrier wafer combination have benefits in addition to improved TTV, improved thermal stability, and improved mechanical stability. For example, costs of assembly can be reduced through the reuse of the silicon carrier wafers. In some instances, a cleaning process (e.g., an etching process) can be used to remove any residual material. A new debond film can then be deposited over the cleaned carrier substrate.

[0021] Referring now to FIG. 1A, a cross-sectional illustration of a carrier **110** is shown, in accordance with an embodiment. In an embodiment, the carrier **110** comprises a substrate **101** and a debond film **105** over the substrate **101**. In an embodiment, the substrate **101** may have a wafer form factor. For example, a diameter of the substrate **101** may be approximately 200 mm, approximately 300 mm, or approximately 450 mm. Though, other wafer sizes may also be used in some embodiments.

[0022] In an embodiment, the substrate **101** comprises a silicon substrate. The silicon substrate **101** may be a high purity silicon material with a low dopant concentration (or substantially no dopants). In an embodiment, the silicon substrate **101** may have a thermal oxide (not shown). The silicon substrate **101** may have any suitable crystal orientation. For example, substrate **101** may have a (100) crystal orientation, a (111) crystal orientation, or a (110) crystal orientation. In an embodiment, the substrate **101** is a material that is substantially transparent to IR radiation. For example, IR radiation between approximately 1 μm and approximately 20 μm may pass through the substrate **101** without significant absorption of the IR radiation. Silicon substrates **101** satisfy this parameter in many embodiments. While silicon may be one example of a suitable substrate **101**, other materials that are at least partially transparent to IR radiation may also be used in some embodiments. For example, the substrate **101** may also comprise germanium, sapphire, or silicon and carbon (e.g., silicon carbide), or glass. As used herein, “partially transparent to IR radiation” may refer to a material that absorbs up to 75% of the IR radiation that passes through the material, absorbs up to 25% of the IR radiation that passes through the material, absorbs up to 10% of the IR radiation that passes through the material, absorbs up to 5% of the IR radiation that passes through the material, or absorbs up to 1% of the IR radiation that passes through the material.

[0023] In an embodiment, the debond film **105** is provided over a top surface of the substrate **101**. The debond film **105** may be a material that is configured to absorb the IR radiation that passes through the substrate **101** during a debonding process. In an embodiment, the debond film **105** may have a thickness T. The thickness T may be up to approximately 500 nm, up to approximately 100 nm, up to approximately 50 nm, up to approximately 20 nm, or up to approximately 10 nm. Though, larger thicknesses T may also be used in some embodiments. For example, the thickness T may be up to approximately 5 μm or greater in some embodiments.

[0024] In an embodiment, the debond film **105** may be deposited with a blanket deposition process. That is, the debond film **105** may be substantially uniform (with respect to thickness and material composition) across the surface of the substrate **101**. For example, deposition processes may comprise PVD, CVD, ALD, and/or the like. In an embodiment, the combination of the debond film **105** and the substrate **101** may have a TTV that is approximately 5 μm or less, or approximately 1 μm or less.

[0025] In an embodiment, the debond film **105** may comprise a material (or materials) that are tuned for absorbing the IR radiation. In an embodiment, the debond film **105** may comprise an inorganic material. For example, the debond film **105** may comprise a thin metal layer or multiple metal layers (e.g., aluminum (Al), tungsten (W), copper (Cu), tantalum (Ta), titanium (Ti),

molybdenum (Mo), manganese (Mn), hafnium (Hf), chromium (Cr), niobium (Nb), zirconium (Zr), palladium (Pd), platinum (Pt), gold (Au), silver (Ag), ruthenium (Ru), tantalum nitride (Ta_N), titanium nitride (Ti_N), niobium nitride (Nb_N), chromium nitride (Cr_N), hafnium nitride (Hf_N), zirconium nitride (Zr_N)). In some embodiments, the debond film **105** may additionally or alternatively include one or more layers of dielectric materials (e.g., silicon dioxide (SiO₂), silicon nitride (Si₃N₄), silicon oxynitride (SiON), silicon carbide (SiC), silicon carbon nitride (SiCN), silicon oxycarbon nitride (SiOCN), aluminum oxide (Al₂O₃), low-k dielectrics such as carbon-doped oxide (CDO) or porous silicon dioxide (SiO₂), titanium dioxide (TiO₂), tantalum oxide (Ta₂O₅), hafnium oxide (HfO₂), or zirconium oxide (ZrO₂)), which may be used to buffer laser ablation and thermal energy, or control thin film interference or adhesion. Thus, in some embodiments, the release layer(s) may be made of one or more materials that include elements such as aluminum (Al), tungsten (W), copper (Cu), titanium (Ti), tantalum (Ta), niobium (Nb), chromium (Cr), hafnium (Hf), molybdenum (Mo), manganese (Mn), zirconium (Zr), palladium (Pd), platinum (Pt), gold (Au), silver (Ag), ruthenium (Ru), silicon (Si), oxygen (O), nitrogen (N), hydrogen (H), and carbon (C), including, without limitation, any of the materials referenced above.

[0026] In the illustrated embodiment of FIG. **1A**, the debond film **105** is shown as a single layer. However, it is to be appreciated that the debond film **105** may comprise two or more layers. For example, the multiple layers may comprise absorption layers, transmission layers, and reflection layers. The reflection layers may reflect IR radiation back to the absorption layers through transmission layers in order to improve ablation of the debond film **105** during the laser release process. Examples of such multi-layer debond films **105** are shown in FIGS. **1B** and **1C**.

[0027] Referring now to FIG. **1B**, a cross-sectional illustration of a carrier **110** is shown, in accordance with an additional embodiment. In an embodiment, the carrier **110** in FIG. **1B** may be similar to the carrier **110** in FIG. **1A**, with the exception of the construction of the debond film **105**. For example, the debond film **105** may comprise a multi-layer stack, such as one with a first layer **102** and a second layer **103** over the first layer. In an embodiment, the first layer **102** may be an inorganic material that is configured to absorb the IR radiation from a laser release process. The second layer **103** may be a layer configured to reflect a significant portion of the IR radiation. The reflected IR radiation is directed back to the first layer **102**, in order to improve the absorption (and ablation) of the first layer **102**. As such, the release process is made more efficient.

[0028] In an embodiment, the first layer **102** may comprise a metallic layer (e.g., aluminum (Al), tungsten (W), copper (Cu), tantalum (Ta), titanium (Ti), molybdenum (Mo), manganese (Mn), hafnium (Hf), chromium (Cr), niobium (Nb), zirconium (Zr), palladium (Pd), platinum (Pt), gold (Au), silver (Ag), ruthenium (Ru), tantalum nitride (Ta_N), titanium nitride (Ti_N), niobium nitride (Nb_N), chromium nitride (Cr_N), hafnium nitride (Hf_N), or zirconium nitride (Zr_N)). In an embodiment, the second layer **103** may comprise a dielectric, such as silicon dioxide (SiO₂), silicon nitride (Si₃N₄), silicon oxynitride (SiON), silicon carbide (SiC), silicon carbon nitride (SiCN), silicon oxycarbon nitride (SiOCN), aluminum oxide (Al₂O₃), low-k dielectrics such as carbon-doped oxide (CDO) or porous silicon dioxide (SiO₂), titanium dioxide (TiO₂), tantalum oxide (Ta₂O₅), hafnium oxide (HfO₂), or zirconium oxide (ZrO₂)).

[0029] In an embodiment, the thickness of the first layer **102** may be different than a thickness of the second layer **103**. In an embodiment, the first layer **102** may be thinner than the second layer **103**. Making the first layer **102** thinner allows for easier ablation of the first layer **102**.

[0030] Referring now to FIG. **1C**, a cross-sectional illustration of a carrier **110** is shown, in accordance with an additional embodiment. In an embodiment, the carrier **110** in FIG. **1C** is similar to the carrier **110** in FIG. **1B**, with the exception of the debond film **105**. Instead of two layers **102** and **103**, a set of three layers (i.e., a first layer **102**, a second layer **103**, and a third layer **104**) are provided. In an embodiment, the first layer **102** may be an inorganic material that is configured to

absorb the IR radiation from a laser release process. The second layer **103** may be a layer configured to reflect a significant portion of the IR radiation and transmit a significant portion of the IR radiation. The reflected IR radiation is directed back to the first layer **102**. The third layer **104** may be configured to reflect a significant portion of the IR radiation back to the first layer **102**, in order to improve the absorption (and ablation) of the first layer **102**. As such, the release process is made more efficient. In an embodiment, the third layer **104** may also be configured to prevent transmission of a significant portion of the IR radiation. While examples of a two layer debond film **105** (FIG. 1B) and a three layer debond film **105** (FIG. 1C) are shown, it is to be appreciated that the debond film **105** can comprise any number of layers, including layers below the first layer **102** and/or above the third layer **104** in FIG. 1C.

[0031] In one embodiment, the first layer **102** and the fourth layer **104** may be similar to each other (as indicated by the shading). Though, in other embodiments, the first layer **102** may be different than the fourth layer **104**. In an embodiment, the compositions, thicknesses, order, etc. of the various layers may be chosen in order to optimize the ablation of at least a portion of the debond film **105**, and reduce or prevent transmission through the debond film **105**. For example, multiple layers of the debond film **105** may be used to buffer laser ablation and thermal energy, control thin film interference between layers, and/or to improve adhesion between the carrier **110** and a subsequently added structure (not shown in FIGS. 1A-1C).

[0032] Referring now to FIGS. 2A-2D, a series of cross-sectional illustrations depicting a process for assembling an electronic device is shown, in accordance with an embodiment. In the embodiments shown in FIGS. 2A-2D, the carrier **210** is shown with a debond film **205** that is similar to the debond film **105** in FIG. 1A. Though, it is to be appreciated that the debond film **205** may be similar to any of the debond films described in greater detail herein. That is, the debond film **205** may include any number of layers. Further, any debond films described herein may include multiple layer configurations such as those described with respect to FIGS. 1B and 1C.

[0033] Referring now to FIG. 2A, a cross-sectional illustration of a carrier **210** is shown, in accordance with an embodiment. In an embodiment, the carrier **210** may comprise a substrate **201** and a debond film **205**. The substrate **201** may comprise a silicon substrate. The debond film **205** may comprise an inorganic debond film **205** similar to any of the debond films described in greater detail herein. More generally, the carrier **210** may be similar to any of the carriers **110** described in greater detail herein.

[0034] In an embodiment, the carrier **210** has a wafer form factor. The wafer form factor allows for a plurality of devices to be fabricated in parallel with others. However, for simplicity, a single electronic device is shown. As will be appreciated, after debonding, the multiple electronic devices may be singulated before integration into a larger structure or system.

[0035] Referring now to FIG. 2B, a cross-sectional illustration of an assembly **220** is shown, in accordance with an embodiment. The assembly **220** may comprise the carrier **210** and an overlying device **230**. The device **230** may be fabricated up from the carrier **210** in some embodiments. The device **230** may sometimes be referred to as an interposer or the like. In an embodiment, the device **230** may comprise an organic substrate **232**. The organic substrate **232** may comprise organic buildup film, an epoxy, and/or the like. While shown as a monolithic structure in FIG. 2B, it is to be appreciated that the organic substrate **232** may comprise a plurality of laminated layers in some embodiments. Electrical routing (not shown) may also be provided within the organic substrate **232** in some embodiment.

[0036] In an embodiment, one or more dies **235** may be integrated into the device **230**. The dies **235** may be embedded within the organic substrate **232**, provided over the organic substrate **232**, or provided at any other suitable position. In an embodiment, dies **235** may comprise any type of die, such as a processor (e.g., a central processing unit (CPU), a graphics processing unit (GPU), an XPU), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a communications die, a memory die, or the like. Additionally, when two or more dies **235** are

included in the device **230**, the dies **235** may be the same type of die, or the dies **235** may be different types of dies.

[0037] Referring now to FIG. **2C**, a cross-sectional illustration of the assembly **220** during a laser release process is shown, in accordance with an embodiment. As shown, a laser **212** is scanned across the backside surface of the substrate **201** of the carrier **210**. IR radiation **215** passes through the substrate **201** and is absorbed by the debond film **205**. The absorption raises a temperature of the debond film **205** (or a portion of the debond film **205**), and results in the ablation of at least a portion of the debond film **205**. As the laser **212** is scanned across the assembly **220**, the mechanical connection between the device **230** and the carrier **210** is removed so that the device **230** can be lifted off of the carrier **210**.

[0038] Referring now to FIG. **2D**, a cross-sectional illustration of the device **230** after removal from the carrier **210** is shown, in accordance with an embodiment. In an embodiment, the removal of the debond film **205** may leave some layers and/or residue on a surface **231** of the organic substrate **232**. The surface **231** may be cleaned with a cleaning or etching process or the like. In other embodiments, the debond film **205** may be designed such that at least a portion of the debond film **205** is used as a layer on the organic substrate **232**. For example, a portion of the debond film **205** may persist onto the surface **231** of the organic substrate **232**, and the portion of the debond film **205** may be used as a solder resist or the like. As used herein, the remaining portion of the debond film **205** left of the organic substrate **232** may be a complete layer or complete layers (e.g., an inorganic dielectric layer). Providing an inorganic dielectric layer over the metallic layer that will be ablated allows for overlying copper metal traces and the like to be plated on the inorganic dielectric as opposed to the underlying metal layer. Throughout the application, the debonding processes may all comprise a residual portion (e.g., one or more organic dielectric layers) left on the device that is being manufactured. These layers may be maintained (e.g., for use as a solder mask) or removed, depending on the needs of the device.

[0039] Referring now to FIG. **3**, a process flow diagram of a process **370** for forming a device with a carrier substrate and an IR laser release process is shown, in accordance with an embodiment. In an embodiment, the process **370** may be similar to the process shown above with respect to FIGS. **2A-2D**. However, processes similar to process **370** may be used in order to fabricate devices similar to any of the embodiments described in greater detail herein.

[0040] In an embodiment, the process **370** may begin with operation **371**, which comprises assembling a structure over a carrier substrate. In an embodiment, the structure is mechanically coupled to the carrier substrate by a debond film. The structure may be an organic structure such as an interposer or the like. The structure may comprise any number of dies, components, electrical routing and/or the like. In an embodiment, the carrier substrate and the debond film may be similar to any of the carrier substrates and/or debond films described in greater detail herein. For example, the carrier substrate may be a silicon wafer, and the debond film may be an inorganic layer that absorbs IR radiation.

[0041] In an embodiment, the process **370** may continue with operation **372**, which comprises ablating the debond film with a laser with a wavelength in the IR range. The debond film may be completely ablated, or a portion of the debond film may be ablated. In an embodiment, the IR laser is scanned across the backside of the carrier substrate. The IR radiation passes through the substrate and is absorbed by the debond film. The increased temperature of the debond film leads to ablation of the material.

[0042] In an embodiment, the process **370** may continue with operation **373**, which comprises separating the carrier substrate from the structure. In an embodiment, the structure may be able to be lifted off of the carrier substrate after the laser release process. In some embodiments, the structure may be cleaned following removal from the carrier substrate. Though, in other embodiments residual portions of the debond film on the structure may be used as a solder resist or the like.

[0043] In an embodiment, the process **370** may continue with operation **374**, which comprises cleaning the carrier substrate. The carrier substrate may be cleaned with an etching process or the like. The cleaning process may remove residual portions of the debond film. As a result, a clean substrate (e.g., silicon wafer) is provided that can be used in subsequent processes. For example, a new debond film may be added to the substrate in a subsequent processing operation.

[0044] Referring now to FIGS. **4A-4F**, a series of cross-sectional illustrations depicting a process for forming a device **430** with a carrier substrate and IR laser debond process is shown, in accordance with an embodiment. In an embodiment, the device **430** is formed with an interposer first process, followed by die attachment.

[0045] Referring now to FIG. **4A**, a cross-sectional illustration of a carrier **410** is shown, in accordance with an embodiment. In an embodiment, the carrier **410** may comprise a substrate **401** and a debond film **405** over the substrate **401**. The substrate **401** may be a silicon wafer or the like. The substrate **401** may be similar to any of the carrier substrate structures described in greater detail herein. In an embodiment, the debond film **405** may be similar to any of the debond films described in greater detail herein. For example, the debond film **405** may comprise an inorganic material (or materials) suitable for absorbing IR radiation.

[0046] Referring now to FIG. **4B**, a cross-sectional illustration of an assembly **420** is shown, in accordance with an embodiment. In an embodiment, the assembly **420** may comprise a device **430** is that is built on the carrier **410**, assembled on the carrier **410**, attached to the carrier **410**, and/or the like. In an embodiment, the device **430** may comprise an organic interposer **432**, such as a plurality of laminated buildup layers. In an embodiment, electrical routing (e.g., pads **429**, traces **437**, vias **436**, etc.) may be provided within the interposer **432**. In an embodiment, a solder resist **433** or other organic layer may be provided around pads **428** at a top of the interposer **432**.

[0047] Referring now to FIG. **4C**, a cross-sectional illustration of the assembly **420** after dies **435** are added is shown, in accordance with an embodiment. In an embodiment, the dies **435** may be similar to any of the dies described in greater detail herein. The dies **435** may be connected to the pads **428** through a solder **431** and pads **434**. Though, any suitable interconnect architecture, such as a first level interconnect (FLI), solution may be used to connect the dies **435** to the interposer **432**. As shown in FIG. **4C**, traces **437** within the interposer **432** may be used to electrically and communicatively couple the dies **435** together. Though, in other embodiments the dies **435** remain isolated (electrically) from each other. In an embodiment, a mold layer **439**, such as an epoxy, is provided around the dies **435**.

[0048] Referring now to FIG. **4D**, a cross-sectional illustration of the assembly **420** after the assembly is flipped, and a laser ablation process is initiated is shown, in accordance with an embodiment. In an embodiment, the laser ablation process may include scanning a laser **412** across the backside of the substrate **401**. Laser IR radiation **415** passes through the substrate **401** and is absorbed by the debond film **405**. At least a portion of the debond film **405** is ablated, and the device **430** can be removed from the carrier **410**.

[0049] Referring now to FIG. **4E**, a cross-sectional illustration of the carrier **410** after device **430** is released is shown, in accordance with an embodiment. In an embodiment, the top surface **407** of the substrate **401** may be bare, due to the ablation process. In other embodiments, any residual material on the top surface **407** from the debond film **405** may be removed with an etching process or the like. The resulting substrate **401** can then be processed to add a new debond film **405**, and the carrier **410** can be reused.

[0050] Referring now to FIG. **4F**, a cross-sectional illustration of the device **430** after the interposer **432** is attached to a board **440** is shown, in accordance with an embodiment. In an embodiment, the board **440** may be a motherboard, a printed circuit board (PCB), or the like. The board **440** may be coupled to the interposer **432** with any suitable second level interconnect (SLI) structure. For example, solder balls **442** coupled to bumps **441** on the pads **438** can be used in some embodiments. Other embodiments may comprise the use of pins, sockets, or the like.

[0051] In an embodiment, the device **430** may be singulated before being attached to the board **440**. That is, the device **430** may be part of a wafer level assembly system. After the carrier **410** is removed, the interposer may be singulated in order to provide individual devices **430**. Each individual device **430** may then be mounted to a board **440**.

[0052] Referring now to FIGS. 5A-5L, a series of cross-sectional illustrations depicting a process of forming an interposer with zero-misaligned vias is shown, in accordance with an embodiment. As will be appreciated from the process flow, the vias described herein will have essentially no misalignment due to the manufacturing process. Such a structure may be fabricated in conjunction with a carrier that includes an IR debond film, such as those described in greater detail herein.

[0053] Referring now to FIG. 5A, a cross-sectional illustration of an assembly **520** is shown, in accordance with an embodiment. The assembly **520** may comprise a carrier **510**. The carrier **510** may be similar to any of the carriers described in greater detail herein. For example, the carrier **510** may comprise a silicon wafer substrate **501** with an inorganic debond film **505**. In an embodiment, a seed layer **543** may be provided over the debond film **505**. The seed layer **543** may comprise an electrically conductive material suitable for use in plating processes. In an embodiment, a resist layer **545** is provided over the seed layer **543**. The resist layer **545** may be any suitable photoresist material.

[0054] Referring now to FIG. 5B, a cross-sectional illustration of the assembly **520** after the resist layer **545** is patterned is shown, in accordance with an embodiment. In an embodiment, the resist layer **545** may be patterned with a grayscale mask in order to have a first region **544** that is fully exposed and other regions **546** that are not fully exposed. For example, the mask (not shown) used for exposure may have a first region where light is not attenuated (i.e., to form regions **544**), a second region where light is attenuated in a controlled fashion to attenuate a given percentage of the light to provide partial exposure (i.e., to form regions **546**), and a third region where no light passes in order to not alter the resist layer **545** (i.e., the remaining portions of the resist layer **545**). While a positive-tone resist is shown in FIGS. 5A-5L, similar processes may also be implemented using a negative tone resist.

[0055] Referring now to FIG. 5C, a cross-sectional illustration of the assembly **520** after the first region **544** is developed is shown, in accordance with an embodiment. The developing process forms an opening **547** that exposes the underlying seed layer **543**. The other regions **546** are shown as being unaffected by the developing process. However, some embodiments may include a recess in the regions **546**. However, the underlying seed layer **543** is not exposed. As such, subsequent plating will only occur in the opening **547**.

[0056] Referring now to FIG. 5D, a cross-sectional illustration of the assembly **520** after an interconnect element **548** is provided in the opening **547** is shown, in accordance with an embodiment. The interconnect element **548** may refer to a structure such as a pad, a trace, a plane, or the like. In an embodiment, the interconnect element **548** may be plated up from the seed layer **543** using a plating process, such as an electroplating process. In an embodiment, plating is stopped before the top of the interconnect element **548** reaches the top of the resist layer **545**.

[0057] Referring now to FIG. 5E, a cross-sectional illustration of the assembly **520** after the regions **546** are developed to form openings **547** is shown, in accordance with an embodiment. The openings **547** expose the underlying seed layer **543** to enable plating in the openings **547**.

[0058] Referring now to FIG. 5F, a cross-sectional illustration of the assembly **520** after a second plating operation is shown, in accordance with an embodiment. As shown, interconnect elements **548** may be formed in the openings **547**. Additionally, a via **549** may plate up from the top of the first interconnect element **548**. Since the via **549** is plated up and confined by the same hole in the resist layer **545** used to form the underlying interconnect element **548**, there will be no misalignment between the via **549** and the underlying interconnect element **548**.

[0059] Referring now to FIG. 5G, a cross-sectional illustration of the assembly **520** after the resist layer **545** is stripped is shown, in accordance with an embodiment. The resist layer **545** may be

stripped with an ashing process, or any other suitable etching process that is selective to the resist layer **545** over other materials in the assembly **520**.

[0060] Referring now to FIG. **5H**, a cross-sectional illustration of the assembly **520** after a spacer formation process is shown, in accordance with an embodiment. In an embodiment, the spacer formation process may include depositing a conformal spacer layer over all surfaces. An asymmetric etch can then be used in order to remove the spacer layer from horizontal surfaces. As such, spacers **551** may remain along sidewalls of the interconnect element **548** and vias **549**. The spacers **551** may be dielectric material, such as an oxide, a nitride, or the like.

[0061] Referring now to FIG. **5I**, a cross-sectional illustration of the assembly **520** after the seed layer **543** is etched is shown, in accordance with an embodiment. In an embodiment, the seed layer **543** may be etched with a wet etching process. The etching of the seed layer **543** electrically isolates the interconnect element **548** from each other.

[0062] Referring now to FIG. **5J**, a cross-sectional illustration of the assembly **520** after a buildup layer **532** is deposited over and around the interconnect element **548** and via **549** is shown, in accordance with an embodiment. In an embodiment, the buildup layer **532** may be an organic dielectric material, such as a buildup film, an epoxy, a mold material, and/or the like. In an embodiment a top surface of the buildup layer **532** may be recessed to be substantially coplanar with a top surface of the via **549**.

[0063] Referring now to FIG. **5K**, a cross-sectional illustration of the assembly **520** after operations **5A-5J** are repeated to form a second layer over the first layer is shown, in accordance with an embodiment. Interconnect element **548** may be provided above and below the via **549**. In some embodiments, a seed layer **543** may be provided between the via **549** and the overlying interconnect element **548**. In the case where vias are not desired, interconnect element **548** may be separated from interconnect element **548** of another layer by portions of the buildup layer **532**. The process can be repeated any number of times to provide a desired number of electrical routing layers for the assembly **520**.

[0064] Referring now to FIG. **5L**, a cross-sectional illustration of the assembly **520** during a laser ablation process is shown, in accordance with an embodiment. In an embodiment, the laser ablation process may include scanning a laser **512** across the backside of the substrate **501**. Laser IR radiation **515** passes through the substrate **501** and is absorbed by the debond film **505**. At least a portion of the debond film **505** is ablated, and the buildup layer **532** (with the embedded electrical routing) can be removed from the carrier **510**. In an embodiment, the carrier **510** can be cleaned and reused, similar to other embodiments described in greater detail herein.

[0065] Referring now to FIGS. **6A-6D**, a series of cross-sectional illustrations depicting a die-first assembly process using a carrier with a debond film compatible with IR laser ablation is shown, in accordance with an embodiment. That is to say, embodiments are not limited to the processing flow being die-first or die-last.

[0066] Referring now to FIG. **6A**, a cross-sectional illustration of an assembly **620** is shown, in accordance with an embodiment. In an embodiment, the assembly **620** comprises a carrier **610**. The carrier **610** may be similar to any of the carriers described in greater detail herein. For example, the carrier **610** may comprise a substrate **601** (e.g., a silicon substrate **601**), and a debond film **605** (e.g., an inorganic debond film **605**).

[0067] In an embodiment, one or more dies **635** may be attached to the debond film **605**. The dies **635** may be attached to the debond film **605** with an adhesive (not shown), such as a die attach film (DAF) or the like. In other embodiments, the debond film **605** may sufficiently adhere to the dies **635** without the need of another adhesive. In some embodiments, a mold layer **639**, an epoxy, or the like may be provided around the dies **635**. The surface of the mold layer **639** may be recessed back so that surfaces of the dies **635** are exposed.

[0068] Referring now to FIG. **6B**, a cross-sectional illustration of the assembly **620** after an interposer **632** is attached to the dies **635** to form a device **630** is shown, in accordance with an

embodiment. In an embodiment, the interposer **632** may comprise buildup film or the like. Electrically conductive features (e.g., pads **638**, vias **636**, traces **637**, etc.) may be embedded within the interposer **632**. The interposer **632** may be built up over the dies **635** in a layer-by-layer process. Or a completed interposer may be attached over the dies **635**. In an embodiment, conductive features within the interposer **632** may be used to electrically and communicatively couple together two of the dies **635**. For example, traces **637** may provide electrical coupling between dies **635**.

[0069] Referring now to FIG. **6C**, a cross-sectional illustration of the assembly **620** during a laser release process is shown, in accordance with an embodiment. In an embodiment, the laser ablation process may include scanning a laser **612** across the backside of the substrate **601**. Laser IR radiation **615** passes through the substrate **601** and is absorbed by the debond film **605**. At least a portion of the debond film **605** is ablated, and the device **630** can be removed from the carrier **610**. In an embodiment, the carrier **610** can be cleaned and reused, similar to other embodiments described in greater detail herein.

[0070] Referring now to FIG. **6D**, a cross-sectional illustration of the assembly **620** after the device **630** is coupled to a board **640** is shown, in accordance with an embodiment. In an embodiment, the board **640** may be a motherboard, a PCB, or the like. The board **640** may be coupled to the interposer **632** with any suitable SLI structure. For example, solder balls **642** coupled to bumps **641** on the pads **638** can be used in some embodiments. Other embodiments may comprise the use of pins, sockets, or the like. In a particular embodiment, the bumps **641** may comprise a part of a redistribution layer (RDL) that is provided below the interposer **632**. The RDL may comprise the bumps and a solder mask (not shown). In some embodiments, the solder mask may be part of the debond film **605** that remains attached to the interposer **632** after the debonding process.

[0071] In an embodiment, the device **630** may be singulated before being attached to the board **640**. That is, the device **630** may be part of a wafer level assembly system. After the carrier **610** is removed, the interposer may be singulated in order to provide individual devices **630**. Each individual device **630** may then be mounted to a board **640**.

[0072] Referring now to FIGS. **7A** and **7B**, a pair of cross-sectional illustrations depicting a multi-level die structure fabricated on a carrier is shown, in accordance with an embodiment. Particularly, the TTV benefits of silicon carriers with an inorganic debond film allow for improved control of integration. This allows for more complex integrations where dies are provided in multiple levels of the device.

[0073] Referring now to FIG. **7A**, a cross-sectional illustration of an assembly **720** is shown, in accordance with an embodiment. In an embodiment, the assembly **720** comprises a carrier **710**. The carrier **710** may be similar to any of the carriers described in greater detail herein. For example, the carrier **710** may comprise a substrate **701** (e.g., a silicon substrate **701**) and a debond film **705** (e.g., an inorganic debond film **705**).

[0074] In an embodiment, a device **730** may be fabricated on the carrier **710**. The device **730** may comprise multiple die levels. For example, dies **735A** may be provided in a first level of the device **730**. The dies **735A** may be embedded in a mold layer **739** or the like. An interposer **732** with embedded electrical components and features (e.g., vias **736**, pads **738**, traces **737**, etc.) may be provided over the first level of dies **735A**. In an embodiment, a second level of dies **735B** may be provided over a surface of the interposer **732** opposite from the first level of dies **735A**. The second level of dies **735B** may be coupled to pads **738** by solder **752** or the like. The second level of dies **735B** may also be embedded in a mold layer **739**.

[0075] Referring now to FIG. **7B**, a cross-sectional illustration of the assembly **720** during a laser release process is shown, in accordance with an embodiment. In an embodiment, the laser ablation process may include scanning a laser **712** across the backside of the substrate **701**. Laser IR radiation **715** passes through the substrate **701** and is absorbed by the debond film **705**. At least a portion of the debond film **705** is ablated, and the device **730** can be removed from the carrier **710**.

In an embodiment, the carrier **710** can be cleaned and reused, similar to other embodiments described in greater detail herein. The released device **730** may be singulated (when a wafer-level assembly is used) and attached to a board or the like (not shown).

[0076] Referring now to FIGS. **8A** and **8B**, a pair of cross-sectional illustrations depicting a multi-level die structure fabricated on a carrier is shown, in accordance with an embodiment. Particularly, the TTV benefits of silicon carriers with an inorganic debond film allow for improved control of integration. This allows for more complex integrations where dies are provided in multiple levels of the device.

[0077] Referring now to FIG. **8A**, a cross-sectional illustration of an assembly **820** is shown, in accordance with an embodiment. In an embodiment, the assembly **820** comprises a carrier **810**. The carrier **810** may be similar to any of the carriers described in greater detail herein. For example, the carrier **810** may comprise a substrate **801** (e.g., a silicon substrate **801**) and a debond film **805** (e.g., an inorganic debond film **805**).

[0078] In an embodiment, a device **830** may be fabricated on the carrier **810**. The device **830** may comprise multiple die levels. For example, a die **835A** may be provided in a first level of the device **830**. The die **835A** may be embedded in a mold layer **839** or the like. In an embodiment, electrically conductive pillars **855** may also be provided through a thickness of the mold layer **839** adjacent to the die **835A**. An interposer **832** with embedded electrical components and features (e.g., vias **836**, pads **838**, traces **837**, etc.) may be provided over the first level of dies **835A**. In an embodiment, a second level of dies **835B** may be provided over a surface of the interposer **832** opposite from the first level of dies **835A**. The second level of dies **835B** may be coupled to pads **838** by solder **852** or the like. The second level of dies **835B** may also be embedded in a mold layer **839**.

[0079] Referring now to FIG. **8B**, a cross-sectional illustration of the assembly **820** during a laser release process is shown, in accordance with an embodiment. In an embodiment, the laser ablation process may include scanning a laser **812** across the backside of the substrate **801**. Laser IR radiation **815** passes through the substrate **801** and is absorbed by the debond film **805**. At least a portion of the debond film **805** is ablated, and the device **830** can be removed from the carrier **810**. In an embodiment, the carrier **810** can be cleaned and reused, similar to other embodiments described in greater detail herein. The released device **830** may be singulated (when a wafer-level assembly is used) and attached to a board or the like (not shown).

[0080] FIG. **9** illustrates a computing device **900** in accordance with one implementation of the disclosure. The computing device **900** houses a board **902**. The board **902** may include a number of components, including but not limited to a processor **904** and at least one communication chip **906**. The processor **904** is physically and electrically coupled to the board **902**. In some implementations the at least one communication chip **906** is also physically and electrically coupled to the board **902**. In further implementations, the communication chip **906** is part of the processor **904**.

[0081] These other components include, but are not limited to, volatile memory (e.g., DRAM), non-volatile memory (e.g., ROM), flash memory, a graphics processor, a digital signal processor, a crypto processor, a chipset, an antenna, a display, a touchscreen display, a touchscreen controller, a battery, an audio codec, a video codec, a power amplifier, a global positioning system (GPS) device, a compass, an accelerometer, a gyroscope, a speaker, a camera, and a mass storage device (such as hard disk drive, compact disk (CD), digital versatile disk (DVD), and so forth).

[0082] The communication chip **906** enables wireless communications for the transfer of data to and from the computing device **900**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not. The communication chip **906** may implement any of a number of wireless standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family),

WiMAX (IEEE 802.16 family), IEEE 802.20, long term evolution (LTE), Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, Bluetooth, derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The computing device **900** may include a plurality of communication chips **906**. For instance, a first communication chip **906** may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication chip **906** may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

[0083] The processor **904** of the computing device **900** includes an integrated circuit die packaged within the processor **904**. In some implementations of the disclosure, the integrated circuit die of the processor may be part of an electronic package that is fabricated through the use of a silicon carrier wafer with a debond film that is deactivated with an IR laser, in accordance with embodiments described herein. The term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory.

[0084] The communication chip **906** also includes an integrated circuit die packaged within the communication chip **906**. In accordance with another implementation of the disclosure, the integrated circuit die of the communication chip may be part of an electronic package that is fabricated through the use of a silicon carrier wafer with a debond film that is deactivated with an IR laser, in accordance with embodiments described herein.

[0085] In an embodiment, the computing device **900** may be part of any apparatus. For example, the computing device may be part of a personal computer, a server, a mobile device, a tablet, an automobile, or the like. That is, the computing device **900** is not limited to being used for any particular type of system, and the computing device **900** may be included in any apparatus that may benefit from computing functionality.

[0086] The above description of illustrated implementations of the disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. While specific implementations of, and examples for, the disclosure are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize.

[0087] These modifications may be made to the disclosure in light of the above detailed description. The terms used in the following claims should not be construed to limit the disclosure to the specific implementations disclosed in the specification and the claims. Rather, the scope of the disclosure is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

[0088] Example 1: a method of assembling an interposer, comprising: assembling a structure over a carrier substrate, wherein the structure is mechanically coupled to the carrier substrate by a debond film, wherein the structure comprises an organic dielectric layer, and wherein the debond film comprises an inorganic layer; ablating at least a portion of the debond film with a laser, wherein a wavelength of the laser is within an infrared range; and separating the carrier substrate from the structure.

[0089] Example 2: the method of Example 1, wherein the carrier substrate comprises a silicon wafer, and wherein the laser passes through a thickness of the silicon wafer to reach the debond film.

[0090] Example 3: the method of Example 1 or Example 2, wherein the structure comprises: a die embedded in the organic dielectric layer.

[0091] Example 4: the method of Example 3, wherein the structure further comprises: a second die above the die.

[0092] Example 5: the method of Examples 1-4, wherein the structure comprises, a zero-misaligned via.

[0093] Example 6: the method of Examples 1-5, wherein the structure comprises a pillar through at

least a portion of a thickness of the organic dielectric layer.

[0094] Example 7: the method of Examples 1-6, wherein the structure comprises a redistribution layer, and wherein the redistribution layer comprises a solder mask that is originally part of the debond film.

[0095] Example 8: the method of Examples 1-7, wherein the debond film comprises one or more of aluminum (Al), tungsten (W), copper (Cu), titanium (Ti), tantalum (Ta), niobium (Nb), chromium (Cr), hafnium (Hf), molybdenum (Mo), manganese (Mn), zirconium (Zr), palladium (Pd), platinum (Pt), gold (Au), silver (Ag), or ruthenium (Ru).

[0096] Example 9: the method of Examples 1-8, wherein the debond film comprises a metallic layer and an inorganic dielectric layer directly on the metallic layer.

[0097] Example 10: a carrier substrate, comprising: a substrate, wherein the substrate comprises silicon; a debond layer over the substrate, wherein the debond layer comprises: a metallic layer on the substrate; and an inorganic dielectric layer over the metallic layer.

[0098] Example 11: the carrier substrate of Example 10, wherein the substrate is a wafer with at least a 200 mm diameter.

[0099] Example 12: the carrier substrate of Example 10 or Example 11, wherein the metallic layer comprises one or more of aluminum (Al), tungsten (W), copper (Cu), titanium (Ti), tantalum (Ta), niobium (Nb), chromium (Cr), hafnium (Hf), molybdenum (Mo), manganese (Mn), zirconium (Zr), palladium (Pd), platinum (Pt), gold (Au), silver (Ag), or ruthenium (Ru).

[0100] Example 13: the carrier substrate of Examples 10-12, wherein the dielectric layer comprises one or more of silicon, oxygen, nitrogen, carbon, aluminum, titanium, hafnium, zirconium, or tantalum.

[0101] Example 14: the carrier substrate of Examples 10-13, wherein the debond layer has a thickness up to 200 nm.

[0102] Example 15: the carrier substrate of Examples 10-14, wherein the substrate comprises an oxide layer over the silicon.

[0103] Example 16: the carrier substrate of Examples 10-15, wherein the silicon has a (100) crystal orientation, a (110) crystal orientation, or a (111) crystal orientation.

[0104] Example 17: a method of forming an electronic device, comprising: assembling an electronic package on a carrier substrate, wherein the electronic package is mechanically coupled to the carrier substrate by an inorganic debond film; debonding the electronic package from the carrier substrate by applying an infrared laser to the debond film to ablate at least a portion of the debond film; and attaching the electronic package to a board.

[0105] Example 18: the method of Example 17, wherein assembling the electronic package on the carrier substrate is a wafer-level assembly process.

[0106] Example 19: the method of Example 17 or Example 18, wherein the debond film comprises one or more of aluminum (Al), tungsten (W), copper (Cu), titanium (Ti), tantalum (Ta), niobium (Nb), chromium (Cr), hafnium (Hf), molybdenum (Mo), manganese (Mn), zirconium (Zr), palladium (Pd), platinum (Pt), gold (Au), silver (Ag), or ruthenium (Ru).

[0107] Example 20: the method of Examples 17-19, wherein the electronic device is part of a personal computer, a server, a mobile device, a tablet, or an automobile.

Claims

1. A method of assembling an interposer, comprising: assembling a structure over a carrier substrate, wherein the structure is mechanically coupled to the carrier substrate by a debond film, wherein the structure comprises an organic dielectric layer, and wherein the debond film comprises an inorganic layer; ablating at least a portion of the debond film with a laser, wherein a wavelength of the laser is within an infrared range; and separating the carrier substrate from the structure.

2. The method of claim 1, wherein the carrier substrate comprises a silicon wafer, and wherein the

laser passes through a thickness of the silicon wafer to reach the debond film.

3. The method of claim 1, wherein the structure comprises: a die embedded in the organic dielectric layer.

4. The method of claim 3, wherein the structure further comprises: a second die above the die.

5. The method of claim 1, wherein the structure comprises a zero-misaligned via.

6. The method of claim 1, wherein the structure comprises a pillar through at least a portion of a thickness of the organic dielectric layer.

7. The method of claim 1, wherein the structure comprises a redistribution layer, and wherein the redistribution layer comprises a solder mask that was part of the debond film.

8. The method of claim 1, wherein the debond film comprises one or more of aluminum (Al), tungsten (W), copper (Cu), titanium (Ti), tantalum (Ta), niobium (Nb), chromium (Cr), hafnium (Hf), molybdenum (Mo), manganese (Mn), zirconium (Zr), palladium (Pd), platinum (Pt), gold (Au), silver (Ag), or ruthenium (Ru).

9. The method of claim 1, wherein the debond film comprises a metallic layer and an inorganic dielectric layer directly on the metallic layer.

10. A carrier substrate, comprising: a substrate, wherein the substrate comprises silicon; a debond layer over the substrate, wherein the debond layer comprises: a metallic layer on the substrate; and an inorganic dielectric layer over the metallic layer.

11. The carrier substrate of claim 10, wherein the substrate is a wafer with at least a 200 mm diameter.

12. The carrier substrate of claim 10, wherein the metallic layer comprises one or more of aluminum (Al), tungsten (W), copper (Cu), titanium (Ti), tantalum (Ta), niobium (Nb), chromium (Cr), hafnium (Hf), molybdenum (Mo), manganese (Mn), zirconium (Zr), palladium (Pd), platinum (Pt), gold (Au), silver (Ag), or ruthenium (Ru).

13. The carrier substrate of claim 10, wherein the dielectric layer comprises one or more of silicon, oxygen, nitrogen, carbon, aluminum, titanium, hafnium, zirconium, or tantalum.

14. The carrier substrate of claim 10, wherein the debond layer has a thickness up to 200 nm.

15. The carrier substrate of claim 10, wherein the substrate comprises an oxide layer over the silicon.

16. The carrier substrate of claim 10, wherein the silicon has a (100) crystal orientation, a (110) crystal orientation, or a (111) crystal orientation.

17. A method of forming an electronic device, comprising: assembling an electronic package on a carrier substrate, wherein the electronic package is mechanically coupled to the carrier substrate by an inorganic debond film; debonding the electronic package from the carrier substrate by applying an infrared laser to the debond film to ablate at least a portion of the debond film; and attaching the electronic package to a board.

18. The method of claim 17, wherein assembling the electronic package on the carrier substrate is a wafer-level assembly process.

19. The method of claim 17, wherein the debond film comprises one or more of aluminum (Al), tungsten (W), copper (Cu), titanium (Ti), tantalum (Ta), niobium (Nb), chromium (Cr), hafnium (Hf), molybdenum (Mo), manganese (Mn), zirconium (Zr), palladium (Pd), platinum (Pt), gold (Au), silver (Ag), or ruthenium (Ru).

20. The method of claim 17, wherein the electronic device is part of a personal computer, a server, a mobile device, a tablet, or an automobile.
